

Olist Brazilian E-Commerce Dataset Analysis

(Customer Acquisition, Retention & Lifecycle Marketing Insights)

1. Summary / Introduction:

I was analysis on an e-commerce dataset (Olist Brazilian E-commerce dataset) and I find the actionable business insights for the marketing and growth teams. The analysis focused on customer behavior, delivery performance, payment patterns, customer satisfaction, and retention opportunities.

The primary business goals were:

- Understand factors affecting customer satisfaction
- Identify customer purchasing behavior and retention patterns
- Analyze delivery performance and operational bottlenecks
- Discover high-value customer segments .

I use python (Pandas, Matplotlib, Seaborn, Plotly, scipy), power BI, and statistical hypothesis testing.

Dataset Overview :

Metric	value
Total Orders	99441
Total Customers	99441
Total Revanue	16008977.41
Average Order Value	160.989
Average Review Score	4.086 %
Delayed Deliver %	7.87%

2. Exploratory Data Analysis (EDA):

A. Order & Revenue Trends:

discussion points:

- Growth trend
- Seasonality
- Sudden spikes or drops
- Business implications

B. Customer Behavior Analysis

discussion points:

- Repeat customers
- High-value customer behavior
- Customer distribution
- Order frequency

C. Delivery Performance Analysis

discussion points:

- Delayed delivery percentage
- Shipping duration
- Regional delivery differences

D. Customer Satisfaction Analysis:

discussion points:

- Review score distribution
- Delayed vs non-delayed orders

E. Payment Behavior Analysis:

discussion points:

- Most used payment methods
- High-value payment types
- Installment usage

3. Hypotheses and Statistical Testing:-

Hypothesis 1:

Question:

Null Hypothesis (H0) = Delivery delays do NOT affect review scores

Alternative Hypothesis (H1) = Delivery delays DO affect review scores.

Statistical Test Used

T-test

Reason for choosing the test:

-> Because I compare between 2 groups average

Result:

T-Statistic: -108.93232066030252

P-Value: 0.0

I reject the null hypothesis because the p-value is exactly 0.0, which falls well below the standard 0.05 significance level. The extremely large negative T-statistic (-108.93) proves that the sample mean is vastly lower than the hypothesized mean, guaranteeing these findings did not happen by random chance. Ultimately, this confirms that late deliveries significantly reduce customer review scores, indicating that delivery performance strongly impacts customer satisfaction and requires operational improvement.

Hypothesis 2:

Question:

h0 = Installment users and non-installment users spend the same amount.

h1 = Installment users spend more money.

Statistical Test Used

T-test

Reason for choosing this test:

-> Because I compare between 2 groups average

Result:

T-Statistic: -55.99263447314089

P-Value: 0.0

Know we reject the null hypothesis because this test show the significant difference in spending behavior between single-payment and installment-payment customers ($p < 0.05$). Customers using installment payments tend to spend significantly more money compared to customers paying in a single transaction. so we learn through this to provide the flexible payment options to encourage customers to purchase higher-value products.

Hypothesis 3:

Question:

H_0 = Payment type and order status are independent.

H_1 = Payment type and order status are related.

Statistical Test Used

Chi-Square Test

Reason for choosing this test:

-> Because I compare between Categories

Result:

Chi-Square Value: 872.0018

P-Value: 1.5113

I reject the null hypothesis because the p-value is exactly 0.0 ($1.51e-165$), which is vastly lower than the standard significance level of 0.05. The Chi-Square value is extremely large (872.00), which guarantees that the relationship between payment type and order status did not happen by random chance. This proves that different payment methods strongly impact order outcomes like successful deliveries, cancellations, or processing issues. The company should definitely analyze which payment methods have higher failure rates to improve transaction efficiency and customer experience

LLM Usage Disclosure:-

ChatGPT (OpenAI) was used for:

- Python syntax assistance
- Debugging dataframe merge issues
- Data cleaning guidance
- Visualization formatting assistance

All insights, hypotheses, interpretations, and recommendations were independently developed based on the dataset analysis.

Files Submitted

- 01_data_cleaning_eda.ipynb
- 02_hypothesis_testing.ipynb
- 03_visual_hebuild_case_study.pbix
- Final Summary Report (PDF)

Git link: https://github.com/adityadubey381/Habuild_Olist_CaseStudy_Aditya